Using Existing Datasets for **TinyML**

Leave space for headshot

Don’t ***collect*** from scratch

Data collection is **difficult**!

- Can we ***reuse*** existing data?

Don’t *collect* from scratch

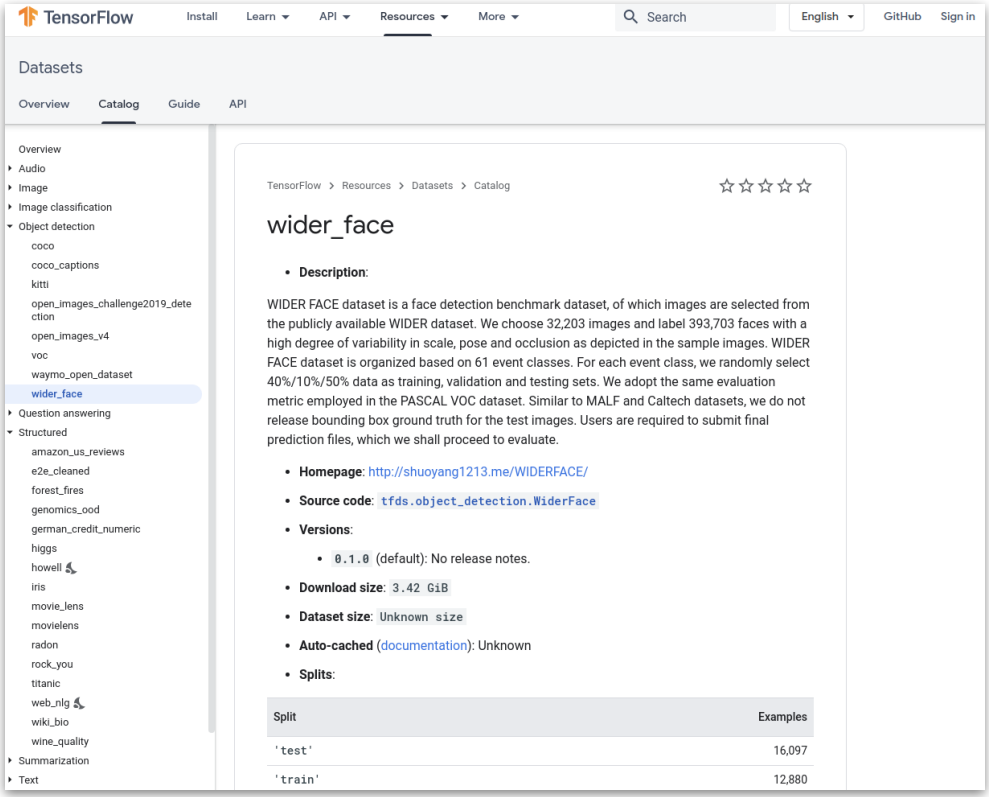
Data collection is **difficult**!

- Can we ***reuse*** existing data?

What’s available?

What’s missing?

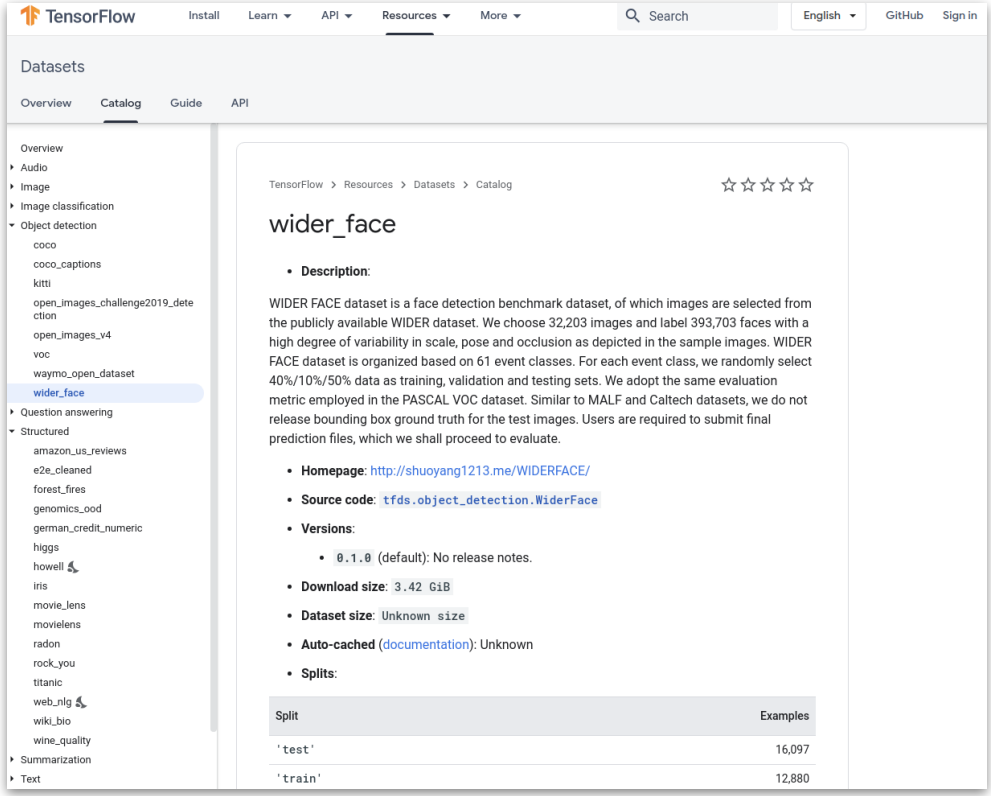
TensorFlow
Datasets Catalog



TensorFlow

Datasets Catalog

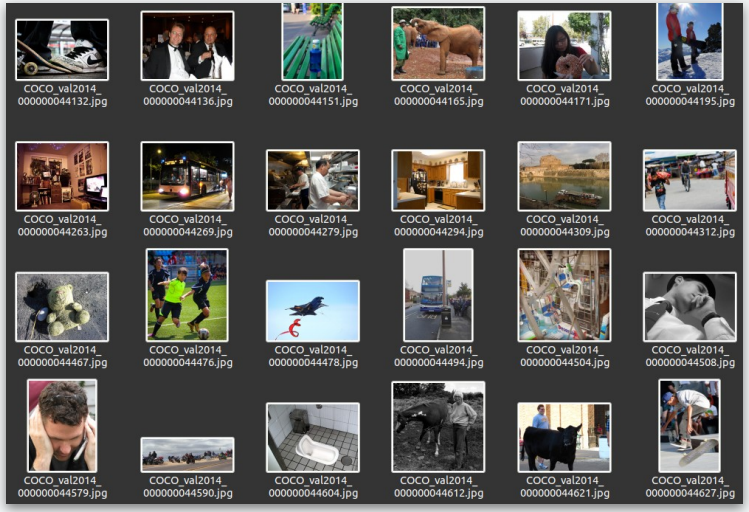
Audio
Image
Image Classification
Object Detection
Question Answering
Structured
Summarization
Text
Translate
Video



TinyML
Person Detection

- **Visual Wake Words**: a new dataset built from Common Objects in Context (**COCO**)
  - ***people*** v. ***no people***

Repurposing existing datasets for **TinyML** tasks is a powerful concept



Don’t *learn* from scratch

- **Transfer** learning

Don’t *learn* from scratch

- **Transfer** learning
- **Pretrained** models: your “AI Data Labeling Assistant”

Don’t *learn* from scratch

- **Transfer** learning
- **Pretrained** models: your “AI Data Labeling Assistant”
- **Generate** your own data
  - Simulations
  - ML models

Fullscreen
Show Presenter